TCS349: Responsible and Explainable AI

Unit 1 | Lectures 8 & 9

Fairness Trade-offs & Bias Mitigation in AI

Comparing equal opportunity, equalized odds and predictive parity; the trade-offs between them; how bias is mitigated before, during and after training; and the risks AI models carry into the real world.

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Metrics & Trade-offs

Bias Mitigation

Imminent Risks

Long-Term Risks

1. Recap: Group Fairness So Far

Demographic parity: the positive-prediction rate should be equal across groups.

Equal opportunity: the True Positive Rate (TPR) should be equal across groups.

Equalized odds: both TPR and FPR should be equal across groups - the strictest of the three.

Today we add a fourth idea, predictive parity, and examine why satisfying every metric at once is rarely possible.

2. Predictive Parity - a Third Fairness Metric

Predictive parity (precision parity): among people predicted positive, the proportion who are truly positive should be equal across groups.

Formally: PPV = TP / (TP + FP) should match across Group A and Group B.

It asks a different question from equalized odds - "When the model says yes, how often is it right, equally for everyone?"

Predictive parity is closely tied to real-world decision quality, since it reflects how reliable a positive prediction actually is.

3. Comparing the Three Metrics

Metric	What It Equalises	Key Question It Answers
Equal Opportunity	TPR across groups	Do qualified people get equal positive outcomes?
Equalized Odds	TPR and FPR across groups	Are correct and incorrect positive predictions both balanced?
Predictive Parity	Precision (PPV) across groups	When the model predicts positive, is it equally reliable for everyone?

4. The Fairness Trade-off

Except in special cases, a model cannot satisfy equalized odds and predictive parity together when groups have different base rates.

This is a mathematical result, not a flaw in any one metric - fairness requires a deliberate choice, not a default answer.

Improving one metric for one group can worsen another metric for the same or a different group.

This is why fairness is treated as a design decision made with stakeholders, not a single number to optimise.

5. Worked Example - Testing the Trade-off

Reusing the Lecture 6 loan numbers - Group A: TP 32, FP 12, FN 8, TN 48; Group B: TP 20, FP 12, FN 20, TN 48:

Metric	Group A	Group B	Equal Across Groups?
TPR (Equal Opportunity)	80%	50%	No — 30-pt gap
FPR	20%	20%	Yes
PPV (Predictive Parity)	72.7%	62.5%	No — ~10-pt gap

6. Choosing Metrics for Context

Criminal justice risk scoring: equalized odds is often prioritised - both false positives and false negatives carry serious consequences.

Healthcare screening: equal opportunity (TPR) is often prioritised, since missing a true positive can be life-threatening.

Lending and credit scoring: predictive parity is often emphasised, since a "yes" decision should be equally reliable for every applicant.

There is no universally correct metric - the choice reflects which type of error is most costly in that context.

7. Lecture 8 Summary

Equal opportunity, equalized odds and predictive parity each formalise a different notion of equal treatment.

These metrics generally cannot all be satisfied at once when groups have different base rates.

Metric choice should reflect the real-world cost of false positives versus false negatives in that application.

Next: how to actually reduce the bias these metrics reveal - bias-mitigation strategies.

8. Bias Mitigation - Three Intervention Points

Bias-mitigation techniques intervene at one of three stages of the ML pipeline:

Stage	When It Intervenes	Example Technique
Pre-processing	Before training, on the data	Reweighing, resampling
In-processing	During training, in the objective	Constrained optimisation, adversarial debiasing
Post-processing	After training, on the outputs	Threshold adjustment, calibration

9. Pre-processing Strategies

Reweighing: assign different sample weights so underrepresented or disadvantaged groups contribute proportionally to training.

Resampling: oversample minority groups or undersample majority groups to balance representation.

Relabeling: adjust training labels near the decision boundary to reduce historical bias encoded in the data.

Advantage: model-agnostic - it works with any downstream algorithm since it only changes the data.

10. In-processing Strategies

Constrained optimisation: add a fairness constraint, such as equalized odds, directly into the training objective.

Adversarial debiasing: train a second "adversary" model that tries to predict the sensitive attribute from the main model's output, and penalise the main model when it succeeds.

Regularisation-based methods: add a fairness penalty term to the loss function alongside the accuracy term.

Advantage: often achieves a better accuracy-fairness balance than pre- or post-processing alone.

11. Post-processing Strategies

Threshold adjustment: use a different decision threshold per subgroup so a chosen fairness metric is equalised.

Calibration: adjust predicted probabilities so they reflect true outcome likelihoods equally well across groups.

Reject-option classification: route uncertain predictions near the decision boundary to human review instead of an automated decision.

Advantage: needs no retraining or access to the original training data - it works directly on model outputs.

12. Imminent Risks from AI Models

Discriminatory outcomes: biased predictions can deny people credit, jobs or services they qualify for, right now.

Privacy violations: models trained on sensitive data can leak personal information through outputs or targeted attacks.

Misinformation and manipulation: generative AI can produce convincing false content at scale.

Overreliance and deskilling: users may trust automated decisions without adequate human oversight.

13. Long-term Risks from AI Models

Entrenchment of inequality: biased systems deployed at scale can compound disadvantage across generations.

Erosion of accountability: complex automated decision chains can make it unclear who is responsible for a harmful outcome.

Concentration of power: control over powerful AI systems may concentrate economic and social influence among a few actors.

Loss of human agency: increasing delegation of decisions to AI may gradually reduce meaningful human oversight of important choices.